

Add to Array-form of Integer

Understanding:-

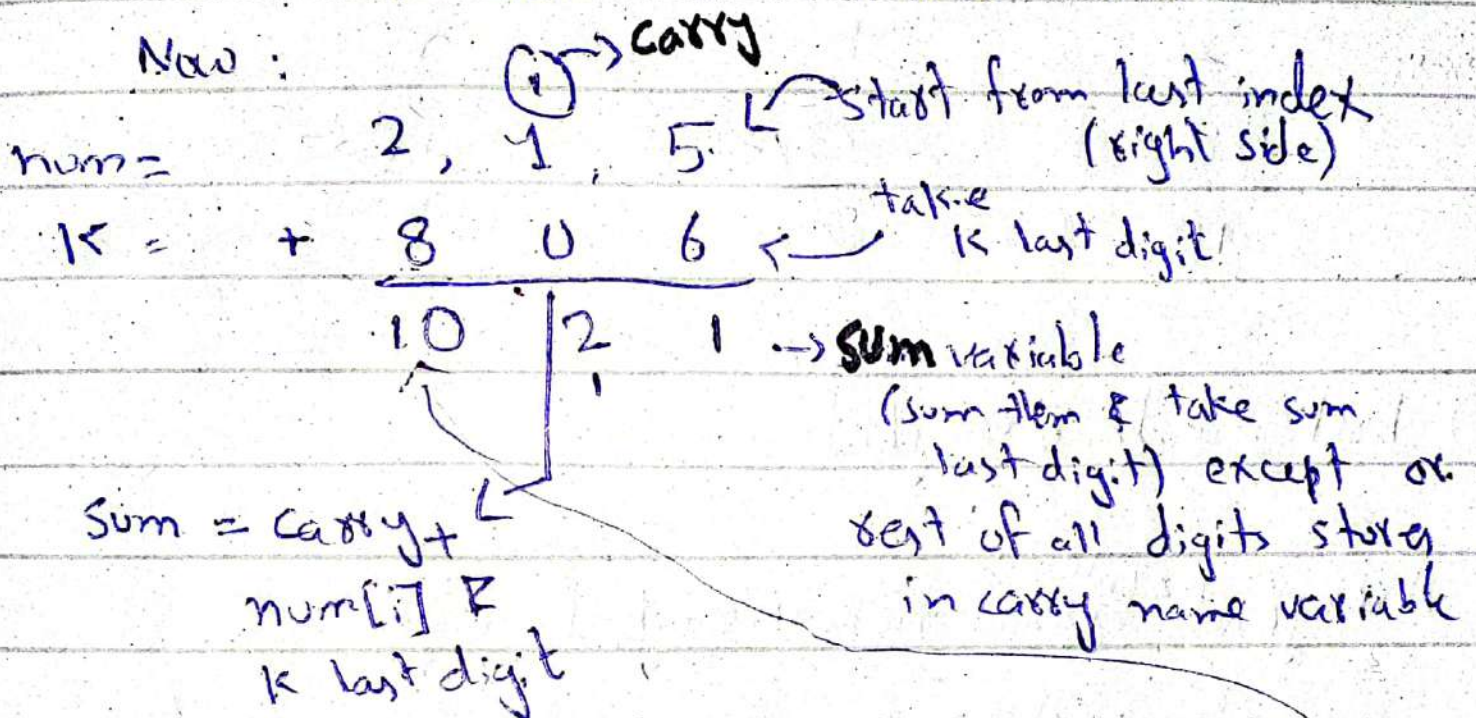
Input:

for example: num [2, 1, 5], k = 806

output: [1, 0, 2, 1]

Explanation: $215 + 806 = 1021$

Now:



Use while loop if nums ends & k is still? or vice versa?

Use ArrayList as a return

for reverse of ans use Collections.reverse()

Also use carry > 0 in while loop after num finishes carry may still left.

Dry Run:

num = [2, 1, 5]

index	0	1	2
	↓	↓	↓
nums	2	1	5
k1	→ 8	0 ←	6 ←
	10	2	1

① First create ArrayList

`ArrayList<Integer> ans = new ArrayList<>();`

② for the starting of index from last digit
`int i = nums.length - 1`

③ Also declare carry variable to store remaining digits.
`int carry = 0`

→ we don't know how many time loop will execute

④ Use while loop → or

while (i >= 0 || carry > 0 || k > 0)

↓
nums. array OR
index > or
equal to zero

↓
after
finish
numbers
carry may
still be left

↓
the k numbers
should be greater
than 0

⑤ make a new variable

then assign num[i] to it

it will store the element value

on num[i] index but

if (i >= 0) index greater or equal to zero

otherwise take x = 0 b/c after finish

of num array may k or carry left to

problem also declare x before if-else condition
to increase scope

int x;

if (i >= 0) { x = num[i]; }

else { x = 0; }

to add

Ex: (1) 2 (1) 4 → num array
+ 9 9 8 → k

(1) 0 2 2
Carry still have/left

(6) Now for taking the last digit of k , create one more variable to store that last digit of k

$$\text{int } y = k \% 10; \text{ (take last digit)}$$

(7) Now create sum variable & sum x variable, y variable & carry variable & store their result in sum variable

$$\text{int } \text{sum} = x + y + \text{carry};$$

(8) Now as sum may have more than 1 digits so in addition we give that numbers ~~to~~ to the left side numbers

ex:
$$\begin{array}{r} 10 \\ + 2 \\ \hline 12 \end{array}$$

So:

$$\begin{array}{r} 10 \\ + 2 \\ \hline 12 \end{array}$$

0 $\rightarrow$ sum is 10 but 1 will store in carry

ans.add(sum/10)

$\downarrow$
store last

digit in arraylist

f for storing the remaining digits of
sum in carry variable

carry = (sum / 10); <sup>remaining
digits
(remove last
digit)</sup>

then i -- ; to move to left
side.

k = k / 10; ~) to remove that last
digit from k

⑨ Now to reverse the ans there is a
called Collections in ArrayList use,

Collections.reverse(ans)

⑩ finally .. return
the ans.